

# The Global Capitalism Index

## Codebook

*Version 1.0*

Democracy and Capitalism Lab, Karsh Institute of Democracy and Institute for Business and  
Society, Darden School of Business  
University of Virginia

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## Suggested Citation

We ask that you attribute our work by citing the **GCI dataset** and the accompanying **white paper**. Please also cite or link to the Global Capitalism Index website at [capitalismindex.com](https://capitalismindex.com).

### GCI Dataset:

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### White Paper:

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### GCI Website:

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## Cautionary Note

- This codebook documents the datasets, transformations, and reliability assessments underlying each of the eight subindices of the Global Capitalism Index (GCI). It is a reference document intended for researchers, policymakers, journalists, and practitioners who use the GCI data. It is not a substitute for the accompanying white paper, which sets out the theoretical foundations, methodological design, and empirical results of the index in full.
- GCI scores are estimates derived from adaptive principal component analysis with reliability adjustment and bootstrap uncertainty quantification. Every country-year score is accompanied by a 95% bootstrap confidence interval. Users should incorporate confidence intervals into any comparative analysis. Differences between countries whose intervals overlap substantially cannot be reliably ranked on the basis of point estimates alone.
- All datasets included in the GCI are drawn from publicly accessible sources, including international organizations, academic publications, and private sector data providers. Every dataset is documented in this codebook with a hyperlink to its original source. Users of GCI scores should also acknowledge the underlying data providers where appropriate.
- The GCI will be updated annually as new data become available. This codebook is versioned to correspond with the associated dataset release. Users should verify they are working with the version of the codebook matching their dataset.
- Reports of documentation errors, dataset issues, or methodological questions should be directed to the GCI research team through the project website. Researchers using the GCI in published work are encouraged to notify the team so that we can maintain an accurate record of applications.
- The GCI is a measurement instrument. Interpretations of scores, subindices, and rankings are the responsibility of the user. The authors of this codebook are not responsible for uses of the data that misrepresent the methodology or exceed the scope of the underlying measurements.

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## Preface

This codebook documents the construction of the Global Capitalism Index (GCI). It is organized as follows.

**Section 1** presents the structural overview of the GCI, showing the mapping of domains to subindices. It is intended as a navigational reference for the detailed documentation that follows.

**Sections 2 through 9** document each of the eight subindices in turn. For each subindex, we present the constituent domains, and within each domain, the individual datasets that contribute to the composite score. Every dataset entry includes: (1) a hyperlink to the original data source, (2) a description of the dataset and, where relevant, the exact question asked or measure produced, (3) the native scale of the dataset before transformation; (4) the transformation applied for use in the GCI, and (5) the reliability grade assigned to the dataset following our evaluation procedure (see the Methodology section in the accompanying white paper for the grading rubric).

**Section 10** details the methodology of the World Economic Forum Executive Opinion Survey, which serves as the source for a substantial share of the perception-based indicators used across multiple subindices. Because these indicators are widely used, we provide extended documentation of their collection and aggregation procedures.

**Section 11** documents the parabolic scaling procedure applied to a small number of datasets for which extreme values in either direction (very low or very high) are theoretically inconsistent with a capitalist institutional configuration. This section includes the reference country set used to calibrate the parabolic peak.

For the theoretical justification of the eight subindices, the derivation of the adaptive PCA procedure, the construction of the composite score, and the empirical results of the index, please consult the accompanying white paper.

# 1 Structure of the Global Capitalism Index

The Global Capitalism Index is organized in three layers. Individual datasets aggregate into domains, domains aggregate into one of eight subindices, and the eight subindices aggregate into the composite GCI score. Aggregation at the subindex and composite levels is performed by adaptive principal component analysis rather than by assigned weights, so the contribution of any dataset or domain is determined by the observed covariance structure of the data in that period. Domains are an organizational and interpretive layer rather than a computational one. The justifications for each of the eight subindex concepts are outlined in the Theory section of the accompanying white paper. The following table maps each of those subindices to their constituent domains, with a brief description of each domain's role within the subindex.

| Subindex                                                 | Domain                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Strength of Property Rights and Private Ownership</b> | Intellectual Property Protection | The ability of firms and individuals to protect their intellectual property, measured through both the formal strength of patent and copyright regimes and the protection firms experience in practice. Without enforceable claims on the non-physical products of human and financial investment, the incentive structure that drives innovation does not form.                                                                                                                                                                                                                                                                                               |
|                                                          | Lack of Corruption               | For property rights and rule of law to function effectively, the system needs to operate with universal and impartial application of the law, where larger or more powerful actors are unable to bend the system to their will. This domain captures the de facto dimension of property rights. Formal institutions, laws, and courts must exist for rights to be exercised, but they must also possess the capacity for enforcement free of practical corruption. Judicial corruption indicators are grouped here rather than under Rule of Law because they measure the integrity of enforcement institutions rather than the substance of legal protection. |

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| Subindex | Domain               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | State Stability      | Without a stable and relatively dependable state, the system of property rights is fallible and susceptible to external pressure and risk. This would damage or eradicate the reliability of the system, rendering it unusable and undependable. This domain measures the capacity for stability and also includes a measure of political constraint against discretionary policy change.                                                                                                                                                      |
|          | Political Expression | The public accountability of companies and the ability for in-depth criticism and discourse of business sentiment without retributive legal action, both upside and downside, is crucial to the vibrancy and multi-directionality of markets. This ability for investigators to produce and publish market and political sentiment free of political interference is crucial for an accurate reflection of property value, rights, and market functioning. This domain measures the information environment rather than electoral competition. |
|          | Rule of Law          | The substantive strength and predictability of the legal order through which individuals, firms, and the state claim or dispute rights of ownership. Rule of law is the mechanism by which property rights are enforced in practice. This domain captures constraints on government power, judicial accountability, compliance with court decisions, judicial review, transparency and predictability of law enforcement, and access to justice.                                                                                               |
|          | Contract Enforcement | The ability of actors within a legal system to create and rely on contracts, measured through court efficiency, insolvency proceedings, and rule-based governance assessments. Contract enforcement is what allows property rights to be honored in practice rather than merely enshrined in law.                                                                                                                                                                                                                                              |

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| Subindex                        | Domain             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | Private Ownership  | The extent to which private ownership is both legally permissible and commonly practiced. This signals broader system protection of ownership access, claims, and protections. It also measures the degree of ownership which bridges the gap between de jure and de facto property rights. For two datasets in this domain, we employ a parabolic maximization approach. Please see section 11 for more details.                                                                                                                                                        |
| <b>Market-Supporting Policy</b> | Market Initiatives | Active initiatives to secure and support the free operation of market systems. This domain does not measure state involvement in markets but state policies that protect and buttress market operation largely through survey measures of antitrust effectiveness, public sector support to start-ups, active labor training, government procurement of advanced technologies, and long-term policy vision. It also incorporates physical infrastructure provisions that link centers of physical, financial, and human capital to enable market creation and operation. |

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| Subindex                     | Domain                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Operational and Institutional Quality | The quality, flexibility, and independence of government institutions that sustain markets once created. This includes central bank independence, regulatory quality, and how conducive legal systems are to market operations. This domain measures the capacity of the state to perform ongoing institutional maintenance and recalibration. This domain also contains the Hamilton Indicator, which scores low mean tariff rates combined with high tariff dispersion and then discounts that score by a measure of particularistic versus public-goods policy orientation, to account for rent-seeking in policy formulation. This indicator is detailed further in the Theory section of the accompanying white paper. |
| <b>Labor Market Openness</b> | Labor Market Flexibility              | The ability of both firms and laborers to allocate human capital as they see fit, measured through ability to hire and fire, the character of labor–employer relations, openness to foreign labor, wage-setting mechanisms, and the linkage of pay to productivity.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                              | Labor Protections                     | Freedom from forced labor and from unequal work arrangements, measured through compliance with internationally recognized labor rights, freedom from coerced labor, and gender equity in pay and economic leadership. Arbitrary constraints such as slavery, race- or gender-based discrimination, and unequal legal treatment artificially undervalue human capital by preventing equal bidding in an open market. A labor system that rejects property in one’s own person, impinges on competition, and suppresses entrepreneurial activity is not structurally capitalist regardless of how profitable it is for its beneficiaries.                                                                                     |

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| Subindex                  | Domain               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Labor Availability   | The supply of new entrants to the labor market and the circulation of workers within it, measured through the foreign-born share of the labor force, internal mobility, the ease of filling vacancies including through foreign hiring, and the availability of skilled labor locally. Like firms competing in a consumer market, labor markets require an open supply of entrants with minimal barriers in order to sustain competition. The domain is intended to capture whether workers possess the option to reallocate their human capital, not the absolute level of turnover. Because globally comparable, annually updated measures of migration policy do not exist at the scale this index requires, the domain proxies the conditions under which labor supply can expand using variables that reflect both policy choice and structural economic conditions. |
| <b>Market Competition</b> | Creative Destruction | The dynamics of firm growth and attrition, market concentration, competitive intensity, and market-relevant innovation. Creative destruction is the process that incessantly revolutionizes economic structure from within and is a foundational mechanism of capitalism rather than a byproduct of it. This domain treats firm entry and exit as necessary but insufficient evidence of creative destruction. Changes in firms' relative market positions (measured using YoY GDP growth as a proximate estimate of economy-wide turnover) also matter in capturing capacity for creative destruction. R&D expenditure and university–industry collaboration measure whether there is sufficient sustained research investment that can generate the technological and organizational advances that market turnover depends on.                                          |

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| Subindex                             | Domain                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capital Market Sophistication</b> | Regulatory Analysis                      | The institutions that support competitive activity and protect firms and consumers from market manipulation, measured through governance quality, insolvency resolution, and the adaptability of legal frameworks to new business models. Capitalism presupposes a baseline regulatory infrastructure that preserves the incentives for success while ensuring consequences for failure. The domain evaluates not the quantity of regulation but its quality and orientation.                                                                                                                                                                                                                             |
|                                      | Human Capital Competitiveness            | The individual, entrepreneurial, and consumer dynamics that support market competitiveness, measured through talent competitiveness, innovative capacity, collaboration within and between firms, workforce mobility, and the teaching of digital and technological skills.                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                      | Access and Participation: Infrastructure | Availability and public participation in formal banking and financial networks, measured through bank branch and ATM density, formal saving and borrowing rates, and the perceived ability of firms to raise capital by issuing shares or bonds. Infrastructure is measured separately from transactions so that the two can be compared against each other. Economies with limited infrastructure typically show low transactional engagement, indicating early-stage development. Economies with extensive infrastructure but modest transaction volumes may signal structural bottlenecks, regulatory friction, or underutilization. Economies dense in both are mature, highly intermediated systems. |

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| Subindex | Domain                                 | Description                                                                                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Access and Participation: Transactions | Frequency and breadth of real financial activity among households and firms, measured through access to loans, the adequacy of financial services to business needs, syndicated lending volume and maturity, domestic credit to the private sector, credit to government and state-owned enterprises, and mobile money and mobile/internet banking activity. |
|          | Equity Market Depth and Scope          | The size, diversity, and trading intensity of public equity markets, alongside measures of market functioning and governance: stock market turnover, mutual fund assets relative to GDP, price volatility, and survey assessment of exchange regulation.                                                                                                     |
|          | Bond Market Depth and Scope            | The development and maturity structure of corporate debt markets, measured through issuance volume relative to GDP, average maturity, and the interest rate spread as an indicator of broader lending conditions.                                                                                                                                            |
|          | Venture Capital Availability           | The prevalence and ease of obtaining funding for innovative, high-growth enterprises, measured through survey assessment of how easily start-ups can find equity funding.                                                                                                                                                                                    |
|          | M&A Prevalence                         | The level of corporate consolidation and capital reallocation, measured through cross-border merger and acquisition sales and purchases. Active markets for corporate control indicate that capital can be redeployed from lower- to higher-return uses without the dissolution of the underlying assets.                                                    |

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| Subindex                                         | Domain                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | Liquidity and Resilience | The systemic capacity to withstand shocks and sustain financial stability, measured through reserve adequacy, country risk premia, and liquid liabilities relative to GDP. Liquidity is a precondition for efficient price discovery, financial asset creation, and open market functioning. Markets that lack it are prone to amplified volatility and diminished investor confidence. Resilience entails both the short-run ability to absorb volatility and macroeconomic change and the longer-run capacity to return to normal operation following crisis. Together the two determine whether capital markets can sustain their intermediation function under adverse conditions.                                                                                                                                                                                                                         |
| <b>Depth and Stability of the Banking System</b> | Banking System Depth     | The degree to which access to the banking system extends across the income and firm-size distribution, measured through financial institution access, depth, and efficiency indices, private credit and deposit bank assets relative to GDP, branch density, card and digital payment penetration, mobile money activity, and survey assessment of access to and innovation in financial services. Increased banking penetration in the broader economy enables the economic dynamism and risk-taking that capitalist systems require, and the institutionalization of banking creates an accessible technology through which risk can be pooled and voluntary exchange made more widely available. Banking infrastructure established mechanisms for assessing creditworthiness and solvency, which created trust in impersonal credit markets and extended access beyond individual financial relationships. |

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| Subindex                                 | Domain                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | Banking System Stability                | The capacity of a banking system to absorb shocks and avoid crises, measured through capital adequacy, regulatory capital to risk-weighted assets, asset quality, liquidity coverage, return on assets, liquid assets to short-term liabilities, default probability, and survey assessment of bank soundness and financial system resilience. Because bank operations contain inherent points of fragility, robustness to shocks determines whether the banking system empowers or periodically destroys capitalist enterprise. When banks fail or credit freezes, uncertainty contracts the money supply, consumer demand, and the flow of capital. |
| <b>New Business Formation and Growth</b> | Human Capital                           | A nation's level of education and basic skills, talent development, attraction, and productivity, all of which supply necessary inputs to new business formation. The embedded knowledge, skill, and education of a talent base is itself productive capital, and endogenous growth theory holds that its accumulation through education, experience, and research drives technological progress and firm formation.                                                                                                                                                                                                                                  |
|                                          | Cultural Acceptance of Entrepreneurship | The systemic acceptance and encouragement of entrepreneurship, measured through attitudes toward entrepreneurial failure, perceived opportunities and capabilities, fear of failure, and the social status accorded to successful entrepreneurs. New business formation is a social phenomenon requiring cultural legitimation of risk-taking, profit motivation, and innovation.                                                                                                                                                                                                                                                                     |

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| Subindex | Domain                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Market Access                | The practical ability of entrepreneurs to develop companies domestically and internationally, measured through export market concentration, competitive intensity, market dominance, cluster development, state ownership, logistics performance, taxes on international trade, antitrust effectiveness, and market scale. Once ventures are created, competition and growth require access to a widening network of market systems to scale.                                                                                                                                         |
|          | Access to Startup Capital    | The ease with which entrepreneurs can access formal and informal credit systems, measured through deposit account and borrower penetration, outstanding deposits and loans from commercial banks, credit unions, and microfinance institutions, domestic credit to the private sector, and survey measures of loan access, venture capital availability, minority shareholder protection, and SME financing. Availability of startup capital provides both the early-stage financing that translates ideas into production and a signal of an economy's support for entrepreneurship. |
|          | Government Supporting Policy | The extent to which government policy enables entrepreneurship through business regulation, market initiatives, competition policy, labor regulation, and taxation, measured through regulatory compliance burden, legal recourse against government action and in commercial disputes, distortive effects of taxes and subsidies, non-tariff barriers, hiring flexibility, long-term policy vision, and legal equality in business.                                                                                                                                                  |

*Continued on next page*

| Subindex                                | Domain             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | R&D Transfers      | The extent to which research and development funding exists and translates in practice into new commercial opportunities, measured through gross expenditure on R&D, patent families by origin, university–industry collaboration, company R&D spending, and expert assessment of whether national research reaches small and medium enterprises. Cooperation among industry, research centers, and the state stimulates new ideas and creates avenues for their commercial realization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Free Flow of Goods &amp; Capital</b> | Free Flow of Trade | The ability of goods and services to flow across borders is a necessary component of international trade and openness to broader global markets. This domain also measures levels of trade participation in practice. Though this may be biased towards countries with more developed trade regimes and endowments, the level of participation in international trade, relative to the size of the economy, does provide a useful indicator of willingness to participate in international markets. It is worth noting that trade-to-GDP ratios are systematically lower for large economies, which tend to utilize extensive internal markets more than smaller countries more dependent on cross-border exchange. However, tariff policy, non-tariff barriers, infrastructure investment, and institutional commitments each shape trade participation independently of country and market scale. Additionally, nations with minimized import/export costs and logistics performance demonstrate an easier environment for trade facilitation. |

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| Subindex | Domain                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Free Flow of Capital       | The ability of financial capital to move across borders, measured through capital account openness, total foreign direct investment flows, and regulatory restrictiveness toward foreign investment. Cross-border capital mobility facilitates risk diversification, broadens markets, and creates active uses for stored savings and earnings. The free deployment of capital across borders reduces market distortions, strengthens private ownership, and encourages capital mobilization. The domain measures both the regulatory environment for capital movement and observed flows. |
|          | Free Flow of Human Capital | The ability of labor and skills to move across borders, measured through net migration and the restrictiveness of regulations governing the hiring of foreign workers. Allowing human capital to be allocated toward its most productive uses enables economies to access global talent pools, and the empirical literature on labor mobility finds that both internal and cross-border movement improves human capital allocation, productivity, and innovation.                                                                                                                          |

## 2 Strength of Property Rights and Private Ownership

### 2.1 Domain 1: Intellectual Property Protection

| Dataset                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                         | Scale | Transformation                         | Grade |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| <a href="#">International Property Rights Alliance Patent Index</a> | This index measures the strength of a country's patent laws based on six key criteria: duration of protection, coverage, restrictions, membership in international treaties, enforcement mechanisms, and patent applications. The data source is the International Patent Index (IPI), created by Dr. Walter Park and updated in its 2023 edition in collaboration with the International Property Rights Alliance. | 0–5   | Min–max normalization to 0–10 per year | 4.75  |
| <a href="#">WEF Intellectual Property Protection</a>                | Survey: “To what extent is intellectual property protected?” Measures practical protection as experienced by firms.                                                                                                                                                                                                                                                                                                 | 1–7   | Min–max normalization to 0–10 per year | 3.725 |
| <a href="#">Chamber of Commerce International IP Index</a>          | Evaluates national IP frameworks including patents, copyrights, enforcement institutions, and commercialization ecosystem. CoC annual report is published in spring each year, so values for a given year are sourced from that year's report (i.e. 2023 values come from the Spring 2023 publication).                                                                                                             | 0–100 | Min–max normalization to 0–10 per year | 4.375 |

### 2.2 Domain 2: Lack of Corruption

| Dataset                                        | Description                                                                                                          | Scale | Transformation                         | Grade |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| <a href="#">Transparency International CPI</a> | Perceptions of public sector corruption. The results are given on a scale of 0 (highly corrupt) to 100 (very clean). | 0–100 | Min–max normalization to 0–10 per year | 4.988 |

*Continued on next page*

| Dataset                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Scale                               | Transformation                                                                   | Grade |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------|-------|
| WGI Control of Corruption             | Control of corruption captures perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as “capture” of the state by elites and private interests.                                                                                                                                                                                                                                            | 0–100                               | Used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| V-Dem Judicial Corrupt Decisions      | Frequency of bribes/extrajudicial payments for favorable legal judgments. Asks “how often do individuals or businesses make undocumented extra payments or bribes in order to speed up or delay the process or to obtain a favorable judicial decision?” V-Dem indicator name is v2jucorrdc.                                                                                                                                                                                    | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |
| V-Dem Judicial Purges                 | Political purges of judges undermining judiciary independence. Question asks “judges are sometimes removed from their posts for cause, as when there is strong evidence of corruption; however, some judges are removed arbitrarily, typically for political reasons. With this distinction in mind, please describe the removal of judges that occurred this calendar year.” with 0 being massive arbitrary purges and 4 being no removals. V-Dem indicator name is v2jupurge. | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |
| V-Dem Government Attacks on Judiciary | Measures political attacks designed to weaken judicial institutions. Asks “how often did the government attack the judiciary’s integrity in public?” with 0 being attacks were carried out on a daily or weekly basis and 4 being there were no attacks on the judiciary’s integrity. V-Dem indicator name is v2jupoatck.                                                                                                                                                       | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |

*Continued on next page*

| Dataset                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Scale                               | Transformation                         | Grade |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|-------|
| V-Dem Court Packing            | Politically motivated expansion of courts to influence outcomes. Survey question asks "the size of the judiciary is sometimes increased for very good reasons, as when judges are added to manage an increasing caseload; however, sometimes judges are added purely for political reasons. With this distinction in mind, please describe any increases in the size of the judiciary that occurred this calendar year" with a 0 being massive political increase in judiciary and a 3 being no evidence of increase being politically motivated. V-Dem indicator name is v2jupack. | 0–3 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |
| V-Dem High Court Independence  | Extent to which high courts rule independently of government pressure. Question asks "When the high court in the judicial system is ruling in cases that are salient to the government, how often would you say that it makes decisions that merely reflect government wishes regardless of its sincere view of the legal record?" with a 0 being always and a 4 being never. V-Dem indicator name is v2juhcind.                                                                                                                                                                    | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |
| V-Dem Lower Court Independence | Independence of non-high-court judges in salient cases. Asks "when judges not on the high court are ruling in cases that are salient to the government, how often would you say that their decisions merely reflect government wishes regardless of their sincere view of the legal record?" with a 0 being always and a 4 being never. V-Dem indicator name is v2juncind.                                                                                                                                                                                                          | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |

## 2.3 Domain 3: State Stability

| Dataset                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Scale | Transformation                                                                   | Grade |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------|-------|
| <a href="#">Fragile States Index</a>         | Composite measure of state stability, institutional capacity, violence, and cohesion.                                                                                                                                                                                                                                                                                                                                                                                                 | 0–120 | Reverse direction, Min–max normalization to 0–10 per year                        | 4.613 |
| <a href="#">WGI Political Stability</a>      | Likelihood of political violence or terrorism.                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0–100 | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| <a href="#">WGI Government Effectiveness</a> | Measures quality of bureaucracy, policy implementation, and public services.                                                                                                                                                                                                                                                                                                                                                                                                          | 0–100 | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| <a href="#">Political Constraint Index</a>   | This index measures the feasibility of policy change, i.e. the extent to which a change in the preferences of any one political actor may lead to a change in government policy. The index scores are derived from a simple spatial model and theoretically ranges from 0 to 1, with higher scores indicating more political constraint and thus less feasibility of policy change. We use POLCONV which includes two additional veto points: the judiciary and sub-federal entities. | 0–1   | min–max normalized per year                                                      | 5     |

## 2.4 Domain 4: Political Expression

| Dataset                                                       | Description                                                           | Scale | Transformation                                                                             | Grade |
|---------------------------------------------------------------|-----------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------|-------|
| <a href="#">Reporters Without Borders Press Freedom Index</a> | Freedom for journalists, media independence, information environment. | 0–100 | Reverse direction so 100 is more free in each year, Min–max normalization to 0–10 per year | 4.3   |

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| Dataset                           | Description                                                             | Scale                               | Transformation                                                                   | Grade |
|-----------------------------------|-------------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------|-------|
| WGI Voice and Accountability      | Measures citizen participation, speech freedoms, media freedoms.        | 0–100                               | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| V-Dem Freedom of Expression Index | Extent of freedom to express political opinions and access information. | 0–1 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |

## 2.5 Domain 5: Rule of Law

| Dataset                       | Description                                                                                                   | Scale                               | Transformation                                                                   | Grade |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------|-------|
| WJP Rule of Law Index         | Measures constraints on government powers, absence of corruption, order/security, civil and criminal justice. | 0–1                                 | Min–max normalization to 0–10 per year                                           | 4.45  |
| WGI Rule of Law               | Perceptions of contract enforcement, police, courts, and property rights.                                     | 0–100                               | Used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| V-Dem Judicial Accountability | Degree to which judges are held accountable for misconduct.                                                   | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |
| V-Dem High Court Compliance   | Government compliance with high court decisions.                                                              | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |
| V-Dem Judicial Review of Laws | Whether courts can invalidate government policies on constitutional grounds.                                  | 0–1 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                           | 5     |

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| Dataset                                           | Description                                                     | Scale                               | Transformation                         | Grade |
|---------------------------------------------------|-----------------------------------------------------------------|-------------------------------------|----------------------------------------|-------|
| V-Dem Transparent and Predictable Law Enforcement | Clarity, publicity, stability, and predictability of law.       | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |
| V-Dem Access to Justice                           | Access to justice for men and women across multiple dimensions. | 0–1 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |

## 2.6 Domain 6: Contract Enforcement

| Dataset                         | Description                                                                                                                                                                                                                                                                                                                                      | Scale | Transformation                         | Grade |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| World Bank Enforcing Contracts  | Court efficiency: time, cost, procedures, and judicial quality.                                                                                                                                                                                                                                                                                  | 0–100 | Min–max normalization to 0–10 per year | 4.4   |
| World Bank Resolving Insolvency | Time, cost, and outcomes of insolvency proceedings; recovery rate for creditors.                                                                                                                                                                                                                                                                 | 0–100 | Min–max normalization to 0–10 per year | 4.4   |
| CPIA Rule-Based Governance      | Property rights and rule-based governance assess the extent to which private economic activity is facilitated by an effective legal system and rule-based governance structure in which property and contract rights are reliably respected and enforced (1=low to 6=high). Sourced from Quality of Government site at University of Gothenburg. | 1–6   | Min–max normalization to 0–10 per year | 4.4   |

## 2.7 Domain 7: Private Ownership

| Dataset                                          | Description                                                                                           | Scale                               | Transformation                         | Grade |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|-------|
| V-Dem State Ownership of the Economy             | Survey asks degree to which state owns capital including land.                                        | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |
| Government Final Consumption Expenditure (% GDP) | Government expenditure on goods and services as share of GDP.                                         | GDP%                                | Parabolic transformation               | 5     |
| Gross Fixed Capital Formation                    | Outlays on additions to the fixed assets of the economy plus net changes in the level of inventories. | GDP%                                | Parabolic transformation               | 5     |

## 3 Market-Supporting Policy

### 3.1 Domain 1: Market Initiatives

| Dataset                                             | Description                                                                                                                                                                                                                                                                                                                                                                              | Scale     | Transformation                                        | Grade |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------|-------|
| WEF Quality of Vocational Training                  | WEF Survey: "In your country, how well does vocational education meet the needs of a competitive economy?"                                                                                                                                                                                                                                                                               | 1–7       | Min–max normalization to 0–10 per year                | 3.875 |
| ICT Development Index                               | Measures global ICT connectivity and development across access, use, and skills.                                                                                                                                                                                                                                                                                                         | 0–100     | Min–max normalization to 0–10 per year                | 4.6   |
| WEF Public Sector Support to Start-ups              | Survey measure of extent to which public sector supports the startup ecosystem.                                                                                                                                                                                                                                                                                                          | 1–7       | Min–max normalization to 0–10 per year                | 3.575 |
| WEF Government Long-Term Vision                     | WEF Survey: "In your country, to what extent does the government have a long-term vision in place?"                                                                                                                                                                                                                                                                                      | 1–7       | Min–max normalization to 0–10 per year                | 3.825 |
| WEF Government Procurement of Advanced Technologies | WEF Survey: "In your country, to what extent do government purchasing decisions foster innovation?"                                                                                                                                                                                                                                                                                      | 1–7       | Min–max normalization to 0–10 per year                | 3.425 |
| WEF Government ICT Policy                           | WEF Survey: "To what extent does the government have a clear implementation plan for utilizing ICTs to improve your country's overall competitiveness?"                                                                                                                                                                                                                                  | 1–7       | Min–max normalization to 0–10 per year                | 3.425 |
| WEF Effectiveness of Anti-Trust Policy              | WEF Survey: "In your country, how effective are anti-monopoly policies at ensuring fair competition?"                                                                                                                                                                                                                                                                                    | 1–7       | Min–max normalization to 0–10 per year                | 3.425 |
| IRF Road Density                                    | Road Network Density by Land Area (km roads / square km land area)                                                                                                                                                                                                                                                                                                                       | Unbounded | Log value then min–max normalization to 0–10 per year | 5     |
| UN Liner Shipping Connectivity Index                | Measures a country's integration into global liner shipping networks. Access to global markets depends heavily on maritime connectivity, especially regular shipping services used to import and export manufactured goods. Higher LSCI values indicate better connectivity. The LSCI is generated for all countries that are serviced by regular containerized liner shipping services. | Unbounded | Log value then min–max normalization to 0–10 per year | 5     |

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| Dataset                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                      | Scale     | Transformation                                                                                                                 | Grade |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|-------|
| <a href="#">Air Transport Freight in million ton-km</a> | Air transport freight is the volume of freight, express, and diplomatic bags carried on each flight stage (operation of an aircraft from takeoff to its next landing), measured in metric tons times kilometers traveled. To scale it by country size, we take the World Bank GDP in constant 2015 USD dataset, scale it to measure in units of millions USD, and take the log of the ratio of air freight over GDP. We then scale that to 0-10. | Unbounded | Take air freight and gdp each in millions, take log value of air freight over gdp, then min-max normalization to 0-10 per year | 5     |

### 3.2 Domain 2: Operational and Institutional Quality

| Dataset                                                                      | Description                                                                                                                                                                     | Scale | Transformation                                                                   | Grade |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------|-------|
| <a href="#">WGI Regulatory Quality</a>                                       | Captures perceptions of government's ability to formulate and implement sound policies that promote private sector development.                                                 | 0-100 | Used World Bank's calculated 0-100 governance score, min-max normalized per year | 5     |
| <a href="#">Central Bank Independence Weighted Index</a>                     | Index of central bank independence based on personnel independence, central bank's objectives, policy formulation, and limits on lending.                                       | 0-1   | Min-max normalization to 0-10 per year                                           | 5     |
| <a href="#">WEF Efficiency of Legal Framework in Challenging Regulations</a> | WEF Survey asked: "In your country, how easy is it for private businesses and citizens to challenge government actions through the legal system?"                               | 1-7   | Min-max normalization to 0-10 per year                                           | 3.875 |
| <a href="#">WEF Efficiency of Legal Framework in Settling Disputes</a>       | WEF Survey asked: "In your country, how efficient are the legal and judicial systems for companies in settling disputes?"                                                       | 1-7   | Min-max normalization to 0-10 per year                                           | 4.025 |
| <a href="#">WEF Ease of Compliance with Government Regulation</a>            | WEF Survey asked: "In your country, how easy is it for companies to comply with government regulation and administrative requirements (e.g., permits, reporting, legislation)?" | 1-7   | Min-max normalization to 0-10 per year                                           | 4.075 |

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| Dataset                                  | Description                                                                                                                                                                                                                                                                                    | Scale                                                            | Transformation                                                   | Grade |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|-------|
| WEF Ease of Complying with Tax System    | WEF Survey asked: “In your country, to what extent are taxes easy to comply with?”                                                                                                                                                                                                             | 1–7                                                              | Min–max normalization to 0–10 per year                           | 3.475 |
| WEF Government Responsiveness to Change  | WEF Survey asked: “In your country, to what extent does the government respond effectively to change (e.g. technological changes, societal and demographic trends, security and economic challenges)?”                                                                                         | 1–7                                                              | Min–max normalization to 0–10 per year                           | 3.975 |
| WEF Government Ensuring Policy Stability | WEF Survey asked: “In your country, to what extent does the government ensure a stable policy environment for doing business?”                                                                                                                                                                 | 1–7                                                              | Min–max normalization to 0–10 per year                           | 3.975 |
| V-Dem Public Sector Corruption Indicator | This V-Dem indicator (v2x_pubcorr) measures the extent to which public sector employees grant favors in exchange for bribes, kickbacks, or other material inducements, and how often they steal, embezzle, or misappropriate public funds or other state resources for personal or family use. | 0–1<br>(where 0 is no corruption, ordinal converted to interval) | Directionally flipped and Min–max normalization to 0–10 per year | 5     |

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| Dataset            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scale                                                          | Transformation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Grade |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Hamilton Indicator | This indicator of our own creation assigns positive values to lower mean tariffs rates in combination with higher coefficients of variation values. We use a 75% weight to mean tariff (WTO simple average MFN rate) and 25% weight to coefficient of variation (WTO coefficient of variation MFN). This captures the Hamiltonian ideal of low tariff rates with targeted tariffs used sparingly. We then weight that value by V-Dem's Particularistic vs Public Goods indicator (v2dlencmps) to proxy the potential for rent-seeking or rent-capturing behavior in policy formulation. | Mean<br>Tariff<br>Rate and<br>Coeffi-<br>cient of<br>Variation | Each underlying dataset are min-max normalized to 0-10 per year + combined into a weighted average of $TariffValue = 0.75 * \text{mean} + 0.25 * \text{variation}$ . This value is then weighted by V-Dem's Particularistic vs Public Good indicator (min-max normalized wherein higher values represent higher levels of public good-focused policy and scaled to 0-1) using a mid-point shrink formula: $FinalValue = 5 + (TariffValue - 5) * (VDEM)$ . This adjusts the final Hamilton indicator value toward a neutral midpoint under higher levels of particularism. Note that this version of the Hamilton indicator is not z-scored, but the iteration within Free Flow of Goods and Capital is z-scored. This is to remain consistent with each subindex's data treatment. | 5     |

## 4 Labor Market Openness

### 4.1 Domain 1: Labor Market Flexibility

| Dataset                                     | Description                                                                                                                                                                              | Scale | Transformation                         | Grade |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Hiring and Firing Practices             | Survey question: “In your country, to what extent do regulations allow the hiring and firing of workers?” Measures regulatory flexibility in hiring and dismissing employees.            | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Cooperation in Labor-Employer Relations | Survey question: “In your country, how would you characterize labor-employer relations?” Captures the extent of cooperative vs. confrontational relations between workers and employers. | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Ease of Hiring Foreign Labor            | Survey question: “In your country, to what extent does the labor market allow companies to hire foreign labor?” Measures openness of regulations and practices to foreign workers.       | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Flexibility of Wage Determination       | Survey question: “In your country, how are wages generally set?” Captures the degree to which wages are determined by markets vs. centralized bargaining.                                | 1–7   | Min–max normalization to 0–10 per year | 3.575 |
| WEF Pay and Productivity                    | Survey question: “to what extent does average pay reflect productivity?” Measures alignment of compensation with worker output.                                                          | 1–7   | Min–max normalization to 0–10 per year | 3.875 |

### 4.2 Domain 2: Labor Protections

| Dataset                                       | Description                                                                                                                                                                                                                                                                                                                                                                 | Scale                               | Transformation                         | Grade |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|-------|
| ILO Compliance with Labor Rights              | Index of the level of national compliance with internationally recognized labor rights based on ILO textual sources and national legislation. Captures the overall strength of labor protections in law and practice. ILO dataset titled “SDG indicator 8.8.2 - Level of national compliance with labor rights (freedom of association and collective bargaining) – Annual” | 0–10                                | Min–max normalization to 0–10 per year | 4.75  |
| V-Dem Freedom From Forced Labor               | V-Dem indicator for the extent to which individuals are free from forced labor, including slavery, bonded labor, and other forms of coercive work. (V-Dem ID is “v2xcl_slave”)                                                                                                                                                                                              | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year | 5     |
| WEF Wage Inequality Between Men and Women     | Survey-based measure of the wage gap between women and men for similar work. Higher values indicate less inequality and more equal pay outcomes.                                                                                                                                                                                                                            | 1–7                                 | Min–max normalization to 0–10 per year | 4.075 |
| WEF Inclusion of Women in Economic Leadership | Survey question on the extent to which women can rise to positions of leadership in firms. Captures gender inclusion in senior economic roles.                                                                                                                                                                                                                              | 1–7                                 | Min–max normalization to 0–10 per year | 4.075 |

### 4.3 Domain 3: Labor Availability

| Dataset                                                 | Description                                                                                                                                                                                                                                             | Scale               | Transformation                                                                                                                                                              | Grade |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| ILO Foreign Born Share of Total Labor Force             | Calculated using ILO dataset “Labor force by sex, age and place of birth (thousands) – Annual” (selecting aggregate age 15 plus, total gender) and creating ratio of foreign-born vs total.                                                             | Raw number in 1000s | Ratio calculated by (raw foreign born / total workers) * 100, then min-max normalization to 0–10 per year where higher ratios of foreign-born employees yield higher scores | 5     |
| WEF Internal Labor Mobility                             | Survey question on the extent of internal labor mobility, i.e., how easily workers can move between jobs, sectors, or regions within the country. “To what extent do people move to other parts of the country to pursue professional opportunities?”   | 1–7                 | Min-max normalization to 0–10 per year                                                                                                                                      | 3.675 |
| WEF Fill Vacancies by Hiring Foreign Labor              | Survey question: the ease with which companies can fill vacancies by hiring foreign labor. Captures openness to foreign workers as a response to skills or labor shortages.                                                                             | 1–7                 | Min-max normalization to 0–10 per year                                                                                                                                      | 3.925 |
| WEF Ease of Finding Skilled Labor in Local Labor Market | Survey-based measure capturing how easy it is for firms to find skilled employees in the domestic labor market. “In your country, to what extent can companies find people with the skills required to fill their vacancies in the local labor market?” | 1–7                 | Min-max normalization to 0–10 per year                                                                                                                                      | 4.025 |

## 5 Market Competition

### 5.1 Domain 1: Creative Destruction

| Dataset                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                        | Scale            | Transformation                         | Grade |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------|-------|
| World Bank New Business Density Rate    | This data measures the number of newly registered companies with limited liability, per calendar year. The new business density is defined as the number of newly registered corporations per 1,000 working-age people (those ages 15–64). The units of measurement are private companies with limited liability. The data can help understand the trends in new firm creation across regions and income groups.                   | Per 1000 workers | Min–max normalization to 0–10 per year | 4.61  |
| World Bank Closed Business Density Rate | This data measures the number of deregistered companies with limited liability, per calendar year. The closed business density is defined as the number of deregistered firms per 1,000 working-age people (those ages 15–64). The units of measurement are private companies with limited liability. The data can help understand trends in deregistration of limited liability companies in the database of business registries. | Per 1000 workers | Min–max normalization to 0–10 per year | 4.46  |
| World Bank Total Business Density Rate  | This data measures the total number of registered companies with limited liability, by the end of each year. The total business density is defined as the total number of registered firms per 1,000 working-age people (those ages 15–64). The units of measurement are private companies with limited liability. The data can help understand the trends in the stock of limited liability companies across economies.           | Per 1000 workers | Min–max normalization to 0–10 per year | 4.51  |

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| Dataset                                    | Description                                                                                                                                                                                                                                                                                                                     | Scale                 | Transformation                         | Grade |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------|-------|
| WEF Intensity of Local Competition         | Based on WEF survey data on entrepreneurship and market competition. “In your country, how intense is competition in the local markets?”                                                                                                                                                                                        | 1–7                   | Min–max normalization to 0–10 per year | 3.375 |
| WEF Nature of Competitive Advantage        | Based on WEF survey data on entrepreneurship and market competition. “On what is the competitive advantage of your country’s companies in international markets based? [1 = primarily low-cost labor or natural resources; 7 = primarily unique products and processes]”                                                        | 1–7                   | Min–max normalization to 0–10 per year | 3.375 |
| WEF Attitudes Towards Entrepreneurial Risk | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent is there a culture of taking risks to pursue entrepreneurial projects?”                                                                                                                                                   | 1–7                   | Min–max normalization to 0–10 per year | 3.875 |
| GDP Growth                                 | Gross domestic product is the total income earned through the production of goods and services in an economic territory during an accounting period.                                                                                                                                                                            | YoY % Growth (2015\$) | Min–max normalization to 0–10 per year | 4.85  |
| WEF Extent of Market Dominance             | Based on WEF survey data on entrepreneurship and market competition. “In your country, how do you characterize corporate activity? [1 = Dominated by a few business groups; 7 = Spread among many firms]”                                                                                                                       | 1–7                   | Min–max normalization to 0–10 per year | 3.875 |
| R&D Expenses as a % of GDP                 | Gross domestic expenditures on research and development (R&D), expressed as a percent of GDP. They include both capital and current expenditures in the four main sectors: Business enterprise, Government, Higher education and Private non-profit. R&D covers basic research, applied research, and experimental development. | %GDP                  | Min–max normalization to 0–10 per year | 4.30  |

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| Dataset                                      | Description                                                                                                                                                                                                    | Scale | Transformation                         | Grade |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF University-Industry Collaboration in R&D | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do business and universities collaborate on research and development?”                                   | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Company Spending on R&D                  | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies invest in research and development (R&D)?”                                                  | 1–7   | Min–max normalization to 0–10 per year | 3.375 |
| WEF Companies Embracing Disruptive Ideas     | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies dynamically adapt their business models to embrace risky or disruptive business ideas?”     | 1–7   | Min–max normalization to 0–10 per year | 3.825 |
| WEF Capacity to Launch New Products          | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies turn ideas into commercially successful new products, services or business models?”         | 1–7   | Min–max normalization to 0–10 per year | 3.275 |
| WEF Growth of Innovative Companies           | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do new companies with innovative ideas grow rapidly and can disrupt established firms in their markets?” | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Companies’ Adaptation to Crisis          | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies prepare for and dynamically adapt to crisis?”                                               | 1–7   | Min–max normalization to 0–10 per year | 3.525 |

## 5.2 Domain 2: Regulatory Analysis

| Dataset                      | Description                                                                                                                                                                                                                                                                                               | Scale | Transformation                                                                   | Grade |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------|-------|
| WGI Political Stability      | Measures state stability and presence of violence.                                                                                                                                                                                                                                                        | 0–100 | Used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| WGI Regulatory Quality       | Regulatory Quality captures perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development.                                                                                                                    | 0–100 | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| WGI Rule of Law              | Similarly measures the strength of legal systems and law enforcement.                                                                                                                                                                                                                                     | 0–100 | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| WGI Control of Corruption    | Control of corruption captures perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as “capture” of the state by elites and private interests.                                                                      | 0–100 | Used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| WGI Government Effectiveness | Government effectiveness captures perceptions of the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government’s commitment to such policies. | 0–100 | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |
| WGI Voice and Accountability | Measures freedom of expression and accountability of public officials.                                                                                                                                                                                                                                    | 0–100 | used World Bank’s calculated 0–100 governance score, min–max normalized per year | 5     |

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| Dataset                                                                       | Description                                                                                                                                                                                                                   | Scale | Transformation                                                                    | Grade |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------|-------|
| <a href="#">World Bank Resolving Insolvency</a>                               | Time, cost, and outcomes of insolvency proceedings; recovery rate for creditors.                                                                                                                                              | 0–100 | Linearly scaled using min–max normalization with each year’s respective min + max | 4.4   |
| <a href="#">WEF Legal Framework’s Adaptability to Digital Business Models</a> | Based on WEF survey data on entrepreneurship and market competition. “In your country, how fast is the legal framework of your country adapting to digital business models (e.g e-commerce, sharing economy, fintech, etc.)?” | 1–7   | Min–max normalization to 0–10 per year                                            | 3.275 |

### 5.3 Domain 3: Human Capital Competitiveness

| Dataset                                                    | Description                                                                                                                                                    | Scale | Transformation                         | Grade |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| <a href="#">INSEAD Global Talent Competitiveness Index</a> | The report is a comprehensive annual benchmarking report that measures how countries grow, attract and retain talent, the only index accredited by the EU.     | 0–10  | Min–max normalization to 0–10 per year | 4.75  |
| <a href="#">WEF Capacity for Innovation</a>                | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies have the capacity to innovate?”             | 1–7   | Min–max normalization to 0–10 per year | 3.375 |
| <a href="#">WEF Pay and Productivity</a>                   | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent does average pay reflect productivity?”                  | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| <a href="#">WEF Encouragement to Idea Generation</a>       | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies encourage employees to generate new ideas?” | 1–7   | Min–max normalization to 0–10 per year | 3.275 |

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| Dataset                                                            | Description                                                                                                                                                                                                         | Scale | Transformation                         | Grade |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Collaboration between Companies                                | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies participate in mutually beneficial collaboration on R&D?”                                        | 1–7   | Min–max normalization to 0–10 per year | 3.775 |
| WEF Collaboration within a Company                                 | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do companies nurture an internal culture of open collaboration?”                                              | 1–7   | Min–max normalization to 0–10 per year | 3.625 |
| WEF Internal Labor Mobility                                        | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent do people move to other parts of the country to pursue professional opportunities?”                           | 1–7   | Min–max normalization to 0–10 per year | 3.675 |
| WEF Educational System’s Teaching of Digital and Technology Skills | Based on WEF survey data on entrepreneurship and market competition. “In your country, how well does the current education system meet the skills needs of a competitive economy in Digital and technology skills?” | 1–7   | Min–max normalization to 0–10 per year | 3.625 |

## 6 Capital Market Sophistication

### 6.1 Domain 1: Access and Participation: Infrastructure

| Dataset                                                                | Description                                                                                                                                                                                                                                                                                                                                  | Scale           | Transformation                         | Grade |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------|-------|
| WEF Local Capital Market Access                                        | Based on WEF survey data on capital markets. “In your country, to what extent can companies raise money by issuing shares and/or bonds on the capital market? [1 = not at all; 7 = to a great extent]”                                                                                                                                       | 1–7             | Min–max normalization to 0–10 per year | 3.375 |
| IMF FAS Bank Branches per 100,000 Adults                               | Each country is calculated as: $100,000 \times (\text{reported number of commercial bank branches} / \text{adult population in the reporting country})$ .                                                                                                                                                                                    | Per 100K Adults | Min–max normalization to 0–10 per year | 4.3   |
| IMF FAS ATMs per 100,000 Adults                                        | Each country is calculated as: $100,000 \times (\text{number of ATMs} / \text{adult population in the reporting country})$ . (World Bank Variable ID: GFDD.AI.25)                                                                                                                                                                            | Per 100K Adults | Min–max normalization to 0–10 per year | 4.300 |
| Global Findex Saved at a Financial Institution (% age 15+)             | Percentage of respondents who report saving or setting aside any money by using an account at a formal financial institution such as a bank, credit union, microfinance institution, or cooperative in the past 12 months (% age 15+). (Corresponds to Global Findex variable <code>fin17a.t.a</code> and World Bank Variable ID GFDD.AI.06) | % of Population | Min–max normalization to 0–10 per year | 3.95  |
| Global Findex Borrowed from a Formal Financial Institution (% age 15+) | Percentage of respondents who report borrowing any money from a bank, credit union, microfinance institution, or another financial institution such as a cooperative in the past 12 months (% age 15+). (Corresponds to Global Findex variable <code>fin22a.t.a</code> and World Bank Variable ID GFDD.AI.07)                                | % of Population | Min–max normalization to 0–10 per year | 3.950 |

### 6.2 Domain 2: Access and Participation: Transactions

| Dataset                                       | Description                                                                                                                                                                                                                                                                                                                                                                                           | Scale    | Transformation                                                                    | Grade |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------|-------|
| WEF Ease of Access to Loans                   | Based on WEF survey data on capital markets. “In your country, how easy is it for businesses to obtain a bank loan? [1 = extremely difficult; 7 = extremely easy]”                                                                                                                                                                                                                                    | 1–7      | Min–max normalization to 0–10 per year                                            | 3.375 |
| WEF Financial Services Meeting Business Needs | Based on WEF survey data on capital markets. “In your country, to what extent does the financial sector provide the products and services that meet the needs of businesses? [1 = not at all; 7 = to a great extent]”                                                                                                                                                                                 | 1–7      | Min–max normalization to 0–10 per year                                            | 3.275 |
| Syndicated Loan Issuance Volume to GDP (%)    | Total volume of newly issued syndicated loans by private entities in industries other than finance, holding companies and insurance, divided by GDP in current USD. GDP is from World Development Indicators. (World Bank Variable ID: GFDD.DM.12)                                                                                                                                                    | % of GDP | Min–max normalization to 0–10 per year                                            | 3.85  |
| Syndicated Loan Average Maturity              | Duration of loans weighted by deal volume. Average maturities are constructed in two steps: (i) the maturity of each tranche is weighted by the value of the tranche to estimate the deal-level weighted-average maturity; (ii) deal-level weighted average maturities are weighted again by the total value of the deal to aggregate to the country-year level. (World Bank Variable ID: GFDD.DM.14) | Years    | Min–max normalization to 0–10 per year                                            | 3.850 |
| Domestic Credit to Private Sector (% GDP)     | Financial resources provided to the private sector by financial corporations, such as through loans, purchases of nonequity securities, and trade credits and other accounts receivable, that establish a claim for repayment. Expressed as a percentage of GDP. WB ID: FS.AST.PRVT.GD.ZS.                                                                                                            | % of GDP | Linearly scaled using min–max normalization with each year’s respective min + max | 3.9   |

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| Dataset                                                             | Description                                                                                                                                                                                                                                                                                           | Scale           | Transformation                         | Grade |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------|-------|
| Credit to Government and SOEs to GDP (%)                            | Raw data are from the electronic version of the IMF's International Financial Statistics. Computed as (IFS line 22a + line 22b + line 22c) / GDP, with substitutions from FOSAG, FOSAOG and FOSAON when unavailable. Local currency GDP is from IFS (line NGDP). (World Bank Variable ID: GFDD.EI.08) | % of GDP        | Min-max normalization to 0–10 per year | 3.900 |
| Value of Mobile Money Transactions (% GDP)                          | Total amount of mobile money transactions carried out by customers of all mobile money service providers during the reference year, as a share of GDP.                                                                                                                                                | % of GDP        | Min-max normalization to 0–10 per year | 3.925 |
| Number of Mobile Money Transactions per 1,000 Adults                | Number of mobile money transactions carried out by customers of all mobile money service providers during the reference year, per 1,000 adults.                                                                                                                                                       | Per 1000 Adults | Min-max normalization to 0–10 per year | 3.925 |
| Value of Mobile and Internet Banking Transactions (% GDP)           | Value of mobile and internet banking transactions carried out by resident non-financial corporations and individuals from the household sector during the reference year (commercial banks only), expressed as a percentage of GDP.                                                                   | % of GDP        | Min-max normalization to 0–10 per year | 4.125 |
| Number of Mobile and Internet Banking Transactions per 1,000 Adults | Number of mobile and internet banking transactions carried out by resident non-financial corporations and individuals from the household sector during the reference year (commercial banks only), per 1,000 adults.                                                                                  | Per 1000 Adults | Min-max normalization to 0–10 per year | 4.125 |

### 6.3 Domain 3: Equity Market Depth and Scope

| Dataset                              | Description                                                                                                                                                                                                                            | Scale      | Transformation                                                                                      | Grade |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------|-------|
| Mutual Fund Assets to GDP (%)        | Ratio of assets of mutual funds to GDP. Mutual funds pool money from many investors to purchase securities; data come from sources such as the Investment Company Institute and national sources. (World Bank Variable ID: GFDD.DI.07) | % of GDP   | Min-max normalization to 0–10 per year                                                              | 3.450 |
| Stock Market Turnover Ratio (%)      | Total value of shares traded during the period divided by the average market capitalization for the period.                                                                                                                            | Percentage | Min-max normalization to 0–10 per year where high turnover yields higher score                      | 3.950 |
| Stock Price Volatility               | Average of the 360-day volatility of the national stock market index. (World Bank Variable ID: GFDD.SM.01)                                                                                                                             | Average    | Direction flipped + Min-max normalization to 0–10 per year where low volatility yields higher score | 3.20  |
| WEF Regulation of Security Exchanges | Based on capital markets and securities exchange survey data. “In your country, to what extent do regulators ensure the stability of the financial market? [1 = not at all; 7 = to a great extent]”                                    | 1–7        | Min-max normalization to 0–10 per year                                                              | 3.375 |

## 6.4 Domain 4: Bond Market Depth and Scope

| Dataset              | Description                                                                                                                                                                                  | Scale | Transformation                                                                                      | Grade |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------|-------|
| Interest Rate Spread | Interest rate charged by banks on loans to private sector customers minus the interest rate paid by commercial or similar banks for deposits. Expressed as a percentage. WB ID: FR.INR.LNDP. | %     | Min-max normalization to 0–10 and direction flipped so that lower spreads yield scores closer to 10 | 4.625 |

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| Dataset                                   | Description                                                                                                                                                                                                                                       | Scale    | Transformation                         | Grade |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------|-------|
| Corporate Bond Issuance Volume to GDP (%) | Total volume of newly issued corporate bonds by private entities in industries other than finance, holding companies and insurance, divided by GDP in current USD. GDP is from World Development Indicators. (World Bank Variable ID: GFDD.DM.13) | % of GDP | Min-max normalization to 0–10 per year | 3.925 |
| Corporate Bond Average Maturity (years)   | Duration of bonds weighted by deal volume, aggregated to the country-year level via volume-weighted averages across tranches and deals. (World Bank Variable ID: GFDD.DM.15)                                                                      | Years    | Min-max normalization to 0–10 per year | 3.925 |

## 6.5 Domain 5: Venture Capital Availability

| Dataset                          | Description                                                                                                                                                                                                   | Scale | Transformation                         | Grade |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Venture Capital Availability | Based on capital markets survey data. “In your country, how easy is it for start-up entrepreneurs with innovative but risky projects to obtain equity funding? [1 = Extremely difficult; 7 = Extremely easy]” | 1–7   | Min-max normalization to 0–10 per year | 3.875 |

## 6.6 Domain 6: M&A Prevalence

| Dataset                | Description                                                                          | Scale               | Transformation                         | Grade |
|------------------------|--------------------------------------------------------------------------------------|---------------------|----------------------------------------|-------|
| Cross-border Sales     | M&A: Retrieved via Passport knowledge hub from Euromonitor International and UNCTAD. | Number of Sales     | Min-max normalization to 0–10 per year | 3.675 |
| Cross-border Purchases | M&A: Retrieved via Passport knowledge hub from Euromonitor International and UNCTAD. | Number of Purchases | Min-max normalization to 0–10 per year | 3.625 |

## 6.7 Domain 7: Liquidity and Resilience

| Dataset                                    | Description                                                                                                                                                                                                                                                               | Scale                 | Transformation                                                                                                                                                                                | Grade |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| IMF Assessing Reserve Adequacy (ARA Ratio) | Ratio of reserves to the IMF's ARA metric; a ratio between 1 and 1.5 suggests that the level of reserves is adequate. Adjustments to the baseline metric may be made based on country characteristics and are documented in IMF staff reports.                            | Central value of 1.25 | Transformed using linear distance transformation to scale to 0–10 per each year of data using 1.25 value as peak. This is based on source methodology detailing between 1-1.5 as ideal value. | 4.775 |
| NYU Country Risk Premium Ratings-Based     | Adjusted default spread or rating-based default spread multiplied by bond market volatility.                                                                                                                                                                              | Number                | Direction flipped such that high premium yields lower score, Min-max normalization to 0–10 per year                                                                                           | 4.725 |
| NYU Country Risk Premium CDS-Based         | Sovereign CDS, net of US, multiplied by a volatility multiplier; a CDS-based measure of country risk premium.                                                                                                                                                             | Number                | Direction flipped such that high premium yields lower score, Min-max normalization to 0–10 per year                                                                                           | 4.725 |
| Liquid Liabilities to GDP                  | Ratio of liquid liabilities (broad money, or M3) to GDP. Liquid liabilities include currency and deposits in the central bank, transferable deposits and electronic currency, time and savings deposits, foreign currency deposits, and certain money market instruments. | % of GDP              | Min-max normalization to 0–10 per year                                                                                                                                                        | 4.375 |

## 7 Depth and Stability of the Banking System

### 7.1 Domain 1: Banking System Depth

| Dataset                                                        | Description                                                                                                                            | Scale                  | Transformation                                   | Grade |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------|-------|
| IMF Financial Institutions Depth Index                         | IMF subindex measuring financial depth of banking institutions.                                                                        | 0–1                    | Z-score + Min–max normalization to 0–10 per year | 4.38  |
| IMF Financial Institutions Access Index                        | IMF subindex measuring access to banking institutions.                                                                                 | 0–1                    | Z-score + Min–max normalization to 0–10 per year | 4.38  |
| IMF Financial Institutions Efficiency Index                    | IMF subindex measuring efficiency of banking institutions.                                                                             | 0–1                    | Z-score + Min–max normalization to 0–10 per year | 4.38  |
| Credit Cards per 1,000 Adults                                  | IMF Financial Access Survey measurement of credit cards per 1,000 adults.                                                              | Number per 1000 adults | Z-score + Min–max normalization to 0–10 per year | 4.3   |
| Debit Cards per 1,000 Adults                                   | IMF Financial Access Survey measurement of debit cards per 1,000 adults.                                                               | Number per 1000 adults | Z-score + Min–max normalization to 0–10 per year | 4.30  |
| Made or Received a Digital Payment (% age 15+)                 | Refers to the percentage of adults who made or received a digital payment during the reference year.                                   | % age 15+              | Z-score + Min–max normalization to 0–10 per year | 3.95  |
| Number of Mobile Money Transactions per 1,000 Adults           | Refers to the mobile money transactions carried out by customers of all mobile money service providers during the reference year.      | Number per 1000 adults | Z-score + Min–max normalization to 0–10 per year | 3.93  |
| WEF Access to Financial Services                               | WEF survey asks “In your country, to what extent do all members of the population have sufficient access to financial services?”       | 1–7                    | Z-score + Min–max normalization to 0–10 per year | 3.675 |
| WEF Innovation of Basic Goods and Services: Financial Services | WEF survey asks “In your country, to what extent is the provision of financial services innovative?”                                   | 1–7                    | Z-score + Min–max normalization to 0–10 per year | 3.675 |
| Private Credit by Deposit Money Banks (% of GDP)               | The financial resources provided to the private sector by domestic money banks as a share of GDP. (World Bank Variable ID: GFDD.DI.01) | Scaled to GDP          | Z-score + Min–max normalization to 0–10 per year | 4.38  |
| Deposit Money Bank Assets to GDP (%)                           | Total assets held by deposit money banks as a share of GDP. (World Bank Variable ID: GFDD.DI.02)                                       | Scaled to GDP          | Z-score + Min–max normalization to 0–10 per year | 4.38  |

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| Dataset                                     | Description                                                                          | Scale                     | Transformation                                   | Grade |
|---------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------|-------|
| Commercial Bank Branches per 100,000 Adults | Number of bank branches reported by IMF Financial Access Surveys per 100,000 adults. | Number per 100,000 adults | Z-score + Min-max normalization to 0–10 per year | 4.38  |

## 7.2 Domain 2: Banking System Stability

| Dataset                                         | Description                                                                                                                                                                                                                                                                                                                                                                | Scale                | Transformation                                                                                                                          | Grade |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------|
| WEF Soundness of Banks                          | WEF survey asks “In your country, what is your assessment of the soundness of banks?”                                                                                                                                                                                                                                                                                      | 1–7                  | Z-score + Min-max normalization to 0–10 per year                                                                                        | 3.725 |
| WEF Financial System Resilience                 | WEF survey asks “In your country, to what extent is the financial system able to respond to crises?”                                                                                                                                                                                                                                                                       | 1–7                  | Z-score + Min-max normalization to 0–10 per year                                                                                        | 3.675 |
| World Bank Bank Z-Score                         | Captures the probability of default of a country’s commercial banking system by comparing capitalization and returns with volatility of returns. (World Bank Variable ID: GFDD.SI.01)                                                                                                                                                                                      | Deviations from mean | Z-score + Min-max normalization to 0–10 per year                                                                                        | 5     |
| Bank Capital to Assets Ratio                    | Measure of capital adequacy comparing Tier 1 capital to total balance sheet assets for deposit takers expressed as a fraction of 100.                                                                                                                                                                                                                                      | Ratio                | Z-score + Min-max normalization to 0–10 per year                                                                                        | 5     |
| Bank Regulatory Capital to Risk-Weighted Assets | Ratio of total regulatory capital to risk-weighted assets held for deposit takers expressed as a fraction of 100.                                                                                                                                                                                                                                                          | Ratio                | Z-score + Min-max normalization to 0–10 per year                                                                                        | 5     |
| Bank Non-Performing Loans to Gross Loans        | The indicator measures the proportion of a deposit taker’s loan portfolio that is impaired or at risk of default. It is calculated as the ratio of non-performing loans (NPLs) to total gross loans, where NPLs are defined as loans that are past due by 90 days or more or are otherwise considered unlikely to be repaid in full without the realization of collateral. | % of total loans     | Z-score + direction flipped such that lower levels of non-performing loans yield higher scores + Min-max normalization to 0–10 per year | 5     |

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| Dataset                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scale   | Transformation                                   | Grade |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------------------|-------|
| Bank Liquidity Coverage Ratio           | The LCR is intended to promote resilience to potential liquidity disruptions over a 30 day horizon. The LCR standard is defined by dividing the stock of high quality liquid assets by net cash outflows over a 30-day time period under stressed conditions. The LCR is not a ratio of balance sheet items, but rather the result of a supervisor-prescribed stress scenario. Data is pulled from IMF Financial Soundness Indicators database (ID FSI288_CFSI_PT) expressed as a fraction of 100 using annual data series. | Ratio   | Z-score + Min-max normalization to 0–10 per year | 5     |
| Bank Return on Assets                   | The quotient of net income before taxes and total (financial and non-financial) assets, expressed as a percentage. Ratio expressed as a fraction of 100. Sourced from the IMF FSI database.                                                                                                                                                                                                                                                                                                                                 | Percent | Z-score + Min-max normalization to 0–10 per year | 5     |
| Liquid Assets to Short-Term Liabilities | This IMF indicator is calculated by using the liquid assets as the numerator and short-term liabilities as the denominator. Liquid assets include currency and deposits and other financial assets available on demand or within three months. Broad measures equal core measure plus securities traded in liquid markets that can be converted into cash with minimal change in value. Short-term refers to three months and should be defined on a remaining maturity basis. (World Bank Variable ID: GFDD.SI.06)         | Ratio   | Z-score + Min-max normalization to 0–10 per year | 5     |

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| Dataset                                          | Description                                                                                                                                                                                                                                                                                                                         | Scale                     | Transformation                                   | Grade |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------|-------|
| <a href="#">World Bank 5-Asset Concentration</a> | Assets of five largest banks as a share of total commercial banking assets. Total assets include total earning assets, cash and due from banks, foreclosed real estate, fixed assets, goodwill, other intangibles, current tax assets, deferred tax, discontinued operations and other assets. (World Bank Variable ID: GFDD.OI.06) | % of total banking assets | Z-score + Min-max normalization to 0–10 per year | 5     |

## 8 New Business Formation and Growth

### 8.1 Domain 1: Human Capital

| Dataset                                                 | Description                                                                                                                                                                                                                                                 | Scale                               | Transformation                         | Grade |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------|-------|
| UNESCO Government Expenditure in Education as % of GDP  | General government expenditure on education (current, capital, and transfers) is expressed as a percentage of GDP.                                                                                                                                          | % of GDP                            | Min-max normalization to 0–10 per year | 4.575 |
| UNDP Average Years of Schooling                         | Average number of years adults over 25 years participated in formal education.                                                                                                                                                                              | Average years                       | Min-max normalization to 0–10 per year | 5     |
| UNESCO Tertiary Enrollment Rates                        | Gross enrollment ratio for tertiary school is calculated by dividing the number of students enrolled in tertiary education regardless of age by the population of the age group which officially corresponds to tertiary education, and multiplying by 100. | Ratio                               | Min-max normalization to 0–10 per year | 4.525 |
| V-Dem Educational Equality                              | “To what extent is high quality basic education guaranteed to all, sufficient to enable them to exercise their basic rights as adult citizens?”                                                                                                             | 0–4 (ordinal converted to interval) | Min-max normalization to 0–10 per year | 5     |
| WEF Internal Labor Mobility                             | WEF Survey asked “In your country, to what extent do people move to other parts of the country to pursue professional opportunities? [1 = Not at all; 7 = To a great extent]”                                                                               | 1–7                                 | Min-max normalization to 0–10 per year | 3.675 |
| WEF Ease of Finding Skilled Labor in Local Labor Market | WEF Survey asked “In your country, to what extent can companies find people with the skills required to fill their vacancies in the local labor market? [1 = Not at all; 7 = To a great extent]”                                                            | 1–7                                 | Min-max normalization to 0–10 per year | 4.025 |
| WEF Availability of Training Services                   | “In your country, how available are high-quality, professional training services? [1 = not available at all; 7 = widely available]”                                                                                                                         | 1–7                                 | Min-max normalization to 0–10 per year | 3.875 |

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| Dataset                                                          | Description                                                                                                                                          | Scale | Transformation                         | Grade |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Pay and Productivity                                         | Based on WEF survey data on entrepreneurship and market competition. “In your country, to what extent does average pay reflect productivity?”        | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Educational System to Teach Digital and Technological Skills | “How well does the current education system meet the skills needs of a competitive economy in Digital and technology skills?”                        | 1–7   | Min–max normalization to 0–10 per year | 3.675 |
| WEF Extent of Staff Training, Companies                          | “In your country, to what extent do companies invest in workforce upskilling and reskilling? [1 = Not at all; 7 = To a great extent]”                | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Quality of Primary Education                                 | “In your country, how well does primary education meet the needs of a competitive economy? [1 = Not at all; 7 = To a great extent]”                  | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Quality of Vocational Training                               | “In your country, how well does vocational education meet the needs of a competitive economy? [1 = Not at all; 7 = To a great extent]”               | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Quality of Secondary Education                               | “In your country, how well does secondary non-vocational education meet the needs of a competitive economy? [1 = Not at all; 7 = To a great extent]” | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Quality of Tertiary Education                                | “In your country, how well does tertiary education meet the needs of a competitive economy? [1 = Not at all; 7 = To a great extent]”                 | 1–7   | Min–max normalization to 0–10 per year | 3.875 |

## 8.2 Domain 2: Cultural Acceptance of Entrepreneurship

| Dataset                                      | Description                                                                                                                                 | Scale | Transformation                         | Grade |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Attitudes toward Entrepreneurial Failure | “In your country, to what extent does fear of failure prevent people from starting new businesses? [1 = To a great extent; 7 = Not at all]” | 1–7   | Min–max normalization to 0–10 per year | 3.875 |

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| Dataset                                           | Description                                                                                                                                                                              | Scale      | Transformation                         | Grade |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------|-------|
| GEM Perceived Opportunities Rate                  | Percentage of 18–64 population (individuals involved in any stage of entrepreneurial activity excluded) who see good opportunities to start a firm in the area where they live.          | Percentage | Min–max normalization to 0–10 per year | 3.35  |
| GEM Perceived Capabilities Rate                   | Percentage of 18–64 population (individuals involved in any stage of entrepreneurial activity excluded) who believe they have the required skills and knowledge to start a business.     | Percentage | Min–max normalization to 0–10 per year | 3.35  |
| GEM Fear of Failure Rate                          | Percentage of 18–64 population (individuals involved in any stage of entrepreneurial activity excluded) who indicate that fear of failure would prevent them from setting up a business. | Percentage | Min–max normalization to 0–10 per year | 3.35  |
| GEM High Status to Successful Entrepreneurs Rate  | Percentage of 18–64 population who agree with the statement that in their country, successful entrepreneurs receive high status.                                                         | Percentage | Min–max normalization to 0–10 per year | 3.35  |
| GEM Entrepreneurship as a Good Career Choice Rate | Percentage of 18–64 population who agree with the statement that in their country, most people consider starting a business as a desirable career choice.                                | Percentage | Min–max normalization to 0–10 per year | 3.35  |

### 8.3 Domain 3: Market Access

| Dataset                        | Description                                                                                                                          | Scale | Transformation                                                          | Grade |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------|-------|
| Herfindahl-Hirschman Index     | Measure of the dispersion of trade value across an exporter’s partners; higher values imply greater concentration and dependency.    | 0–1   | Logged + Min–max normalization to 0–10 per year + directionally flipped | 5     |
| WEF Extent of Market Dominance | “In your country, how do you characterize corporate activity? [1 = Dominated by a few business groups; 7 = Spread among many firms]” | 1–7   | Min–max normalization to 0–10 per year                                  | 3.875 |

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| Dataset                                          | Description                                                                                                                                          | Scale                               | Transformation                                                                                       | Grade |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------|-------|
| WEF Intensity of Local Competition               | “In your country, how intense is competition in the local markets? [1 = not intense at all; 7 = extremely intense]”                                  | 1–7                                 | Min–max normalization to 0–10 per year                                                               | 3.375 |
| V-Dem State Ownership of the Economy             | Expert survey: “Does the state own or directly control important sectors of the economy?”                                                            | 0–4 (ordinal converted to interval) | Min–max normalization to 0–10 per year                                                               | 5     |
| Logistics Performance Index                      | Ranks countries on dimensions of trade logistics, including customs, infrastructure, and timeliness of shipments.                                    | 1–5 (5 = best)                      | Min–max normalization to 0–10 per year                                                               | 4.7   |
| WB GDP Constant 2021 International \$ Value, PPP | Raw GDP value proxying for the ability of a company to be “born global” and operate internationally quickly.                                         | Dollar Value in PPP                 | Min–max normalization to 0–10 per year                                                               | 5     |
| IMF Taxes on International Trade (% of revenue)  | Taxes payable when goods cross national/customs frontiers or when services exchange between residents and non-residents, as a percentage of revenue. | % of revenue                        | directional flipped so that lower taxes equals higher score + Min–max normalization to 0–10 per year | 5     |
| WEF Effectiveness of Anti-Trust Policy           | “In your country, how effective are anti-monopoly policies at ensuring fair competition? [1 = not effective at all; 7 = extremely effective]”        | 1–7                                 | Min–max normalization to 0–10 per year                                                               | 3.425 |
| WEF State of Cluster Development                 | “In your country, how widespread are well-developed and deep clusters? [1 = Nonexistent; 7 = Widespread in many fields]”                             | 1–7                                 | Min–max normalization to 0–10 per year                                                               | 4.075 |

## 8.4 Domain 4: Access to Startup Capital

| Dataset                                                               | Description                                                                                                | Scale                  | Transformation                                      | Grade |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------|-------|
| IMF Number of Deposit Accounts with Commercial Banks per 1,000 Adults | Number of deposit accounts with commercial banks per 1,000 adults pulled from IMF Financial Access Survey. | Number per 1000 adults | Log values + Min–max normalization to 0–10 per year | 4     |

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| Dataset                                                                        | Description                                                                                                                                                                             | Scale                  | Transformation                                      | Grade |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------|-------|
| IMF Outstanding Deposits with Commercial Banks (% of GDP)                      | Outstanding deposits with commercial banks as % of GDP.                                                                                                                                 | % of GDP               | Log values + Min-max normalization to 0–10 per year | 4     |
| IMF Outstanding Deposits with Credit Unions and Credit Cooperatives (% of GDP) | Outstanding deposits with credit unions and credit cooperatives.                                                                                                                        | % of GDP               | Min-max normalization to 0–10 per year              | 4     |
| IMF Borrowers from Commercial Banks Per 1000                                   | Refers to the resident non-financial corporations (public and private) and individuals in the household sector that have obtained credit (loans) from commercial banks per 1000 adults. | Number per 1000 adults | Min-max normalization to 0–10 per year              | 4     |
| IMF Outstanding Loans from All Commercial Banks (% of GDP)                     | Outstanding loans % of GDP from all commercial banks as a percent of GDP.                                                                                                               | % of GDP               | Log + Min-max normalization to 0–10 per year        | 4     |
| IMF Outstanding Loans from Credit Unions and Credit Cooperatives (% of GDP)    | Outstanding loans % of GDP from credit unions and coops.                                                                                                                                | % of GDP               | Log + Min-max normalization to 0–10 per year        | 4     |
| IMF Outstanding Loans from All Microfinance Institutions (% of GDP)            | Outstanding loans % of GDP from microfinance institutions.                                                                                                                              | % of GDP               | Log + Min-max normalization to 0–10 per year        | 4     |
| WB Domestic Credit to Private Sector (% of GDP)                                | Domestic credit to private sector refers to financial resources provided to the private sector by financial corporations.                                                               | % of GDP               | Min-max normalization to 0–10 per year              | 5     |
| WEF Ease of Access to Loans                                                    | “In your country, how easy is it for businesses to obtain a bank loan? [1 = extremely difficult; 7 = extremely easy]”                                                                   | 1–7                    | Min-max normalization to 0–10 per year              | 3.375 |
| WEF Venture Capital Availability                                               | “How easy is it for start-up entrepreneurs with innovative but risky projects to obtain equity funding? [1 = Extremely difficult; 7 = Extremely easy]”                                  | 1–7                    | Min-max normalization to 0–10 per year              | 3.875 |

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| Dataset                                 | Description                                                                                                                                 | Scale | Transformation                         | Grade |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Protection of Minority Shareholders | “To what extent are the interests of minority shareholders protected by the legal system? [1 = not protected at all; 7 = fully protected]”  | 1–7   | Min–max normalization to 0–10 per year | 3.375 |
| WEF Financing of SMEs                   | “To what extent can SMEs access the finance they need for operations through the financial sector? [1 = Not at all; 7 = To a great extent]” | 1–7   | Min–max normalization to 0–10 per year | 3.825 |

## 8.5 Domain 5: Government Supporting Policy

| Dataset                                                     | Description                                                                                                                                                                                            | Scale | Transformation                         | Grade |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Hiring and Firing                                       | WEF Survey asked “In your country, to what extent do regulations allow for the flexible hiring and firing of workers? [1 = Not at all; 7 = To a great extent]”                                         | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Ease of Hiring Foreign Labor                            | WEF Survey asked “In your country, how restrictive are regulations related to the hiring of foreign labor? [1 = Highly restrictive; 7 = Not restrictive at all]”                                       | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Distortive Effect of Taxes and Subsidies on Competition | “In your country, to what extent do fiscal measures (subsidies, tax breaks, etc.) distort competition? [1 = Distort to a great extent; 7 = Do not distort at all]”                                     | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Ease of Compliance with Government Regulation           | “In your country, how easy is it for companies to comply with government regulation and administrative requirements (e.g., permits, reporting, legislation)? [1 = Overly-complex; 7 = Extremely easy]” | 1–7   | Min–max normalization to 0–10 per year | 4.075 |
| WEF Growth of Innovative Companies                          | “In your country, to what extent do new companies with innovative ideas grow rapidly and can disrupt established firms? [1 = Not at all; 7 = To a great extent]”                                       | 1–7   | Min–max normalization to 0–10 per year | 3.875 |

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| Dataset                                                      | Description                                                                                                                                                                                                                                                                                                                                        | Scale | Transformation                         | Grade |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|-------|
| WEF Prevalence of Non-Tariff Barriers                        | “To what extent do non-tariff barriers limit the ability of imported goods to compete domestically? [1 = To a great extent; 7 = Not at all]”                                                                                                                                                                                                       | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Government Long-Term Vision                              | “To what extent does the government have a long-term vision in place? [1 = Not at all; 7 = To a great extent]”                                                                                                                                                                                                                                     | 1–7   | Min–max normalization to 0–10 per year | 3.825 |
| WEF Efficiency of Legal Framework in Challenging Regulations | WEF Survey asked “In your country, how easy is it for private businesses and citizens to challenge government actions through the legal system? [1 = Extremely difficult; 7 = Extremely easy]”                                                                                                                                                     | 1–7   | Min–max normalization to 0–10 per year | 3.875 |
| WEF Efficiency of Legal Framework in Settling Disputes       | WEF Survey asked “In your country, how efficient are the legal and judicial systems for companies in settling disputes? [1 = Extremely inefficient; 7 = Extremely efficient]”                                                                                                                                                                      | 1–7   | Min–max normalization to 0–10 per year | 4.025 |
| World Bank Women, Business, and the Law Index                | Women, Business and the Law tracks progress toward legal equality between men and women in 190 economies in areas specific to business and legal systems, including constraints on starting and owning businesses. Data are collected with standardized questionnaires to ensure comparability across economies. We use the aggregate index value. | 0–100 | Min–max normalization to 0–10 per year | 5     |

## 8.6 Domain 6: R&D Transfers

| Dataset                                                       | Description                                                                                                                                             | Scale      | Transformation                                  | Grade |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------|-------|
| GEM R&D Transfer                                              | Expert survey assessment of “the extent to which national research and development will lead to new commercial opportunities and is available to SMEs”. | 1–9        | Min–max normalization to 0–10 per year          | 3.35  |
| WIPO Patent Family by Origin Based on the Earliest Grant Year | Number of patent families by country filed with WIPO.                                                                                                   | Raw number | Logged + Min–max normalization to 0–10 per year | 4     |

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| Dataset                                      | Description                                                                                                                          | Scale    | Transformation                         | Grade |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------|-------|
| World Bank Gross Expenditure on R&D          | Gross domestic expenditures on research and development (R&D), expressed as a percent of GDP.                                        | % of GDP | Min-max normalization to 0–10 per year | 4.5   |
| WEF University-Industry Collaboration in R&D | “To what extent do business and universities collaborate on research and development (R&D)? [1 = Not at all; 7 = To a great extent]” | 1–7      | Min-max normalization to 0–10 per year | 3.875 |
| WEF Company Spending on R&D                  | “To what extent do companies invest in research and development (R&D)? [1 = do not invest at all; 7 = invest heavily in R&D]”        | 1–7      | Min-max normalization to 0–10 per year | 3.375 |

## 9 Free Flow of Goods and Capital

### 9.1 Domain 1: Free Flow of Trade

| Dataset            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scale                                         | Transformation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Grade |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Hamilton Indicator | This indicator of our own creation assigns positive values to lower mean tariffs rates in combination with higher coefficients of variation values. We use a 75% weight to mean tariff (WTO simple average MFN rate) and 25% weight to coefficient of variation (WTO coefficient of variation MFN). This captures the Hamiltonian ideal of low tariff rates with targeted tariffs used sparingly. We then weight that value by V-Dem's Particularistic vs Public Goods indicator (v2dlencmps) to proxy the potential for rent-seeking or rent-capturing behavior in policy formulation. | Mean Tariff Rate and Coefficient of Variation | Z-Score of each underlying dataset are then min-max normalized to 0–10 per year + combined into a weighted average of $TariffValue = 0.75 * \text{mean} + 0.25 * \text{variation}$ . This value is then weighted by V-Dem's Particularistic vs Public Good indicator (min-max normalized wherein higher values represent higher levels of public good-focused policy and scaled to 0-1) using a mid-point shrink formula: $FinalValue = 5 + (TariffValue - 5) * (VDEM)$ . This adjusts the final Hamilton indicator value toward a neutral midpoint under higher levels of particularism. | 5     |

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| Dataset                                      | Description                                                                                                                                                                                                                                                                                                                                                 | Scale                      | Transformation                                                       | Grade |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------|-------|
| BTI Liberalization of Foreign Trade          | This index analyzes and evaluates whether and how developing countries and countries in transition are steering social change toward democracy and a market economy. The survey question is: “To what extent has foreign trade been liberalized?”                                                                                                           | 1–10                       | Z-Score + Min–max normalization to 0–10 per year                     | 3.3   |
| Trade (% of GDP)                             | Trade is the sum of exports and imports of goods and services measured as a share of gross domestic product.                                                                                                                                                                                                                                                | % of GDP                   | Z-Score + Min–max normalization to 0–10 per year                     | 5     |
| Logistics Performance Index                  | The LPI ranks countries on six dimensions of trade – including customs performance, infrastructure quality, and timeliness of shipments. The data used in the ranking comes from a survey of logistics professionals answering questions about foreign countries with which they operate.                                                                   | 1–5 (5 = best performance) | Z-Score + Min–max normalization to 0–10 per year                     | 4.7   |
| Herfindahl–Hirschman Index                   | This indicator is a measure of the dispersion of trade value across an exporter’s partners. A country with a preponderance of trade value concentrated in a very few markets will have an index value close to 1.                                                                                                                                           | 0–1                        | Z-Score + Min–max normalization to 0–10 per year + direction flipped | 5     |
| Harvard Growth Lab Economic Complexity Index | A measure of the knowledge in a society as expressed in the products it makes. The economic complexity of a country is calculated based on the diversity of exports a country produces and their ubiquity, or the number of the countries able to produce them (and those countries’ complexity) using SITC product classification. (column name eci_sitc). | Roughly -3 to +3           | Z-Score + Min–max normalization to 0–10 per year                     | 5     |

## 9.2 Domain 2: Free Flow of Capital

| Dataset                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Scale                      | Transformation                                                           | Grade |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------|-------|
| FDI Total Flows                      | We measure cross-border capital mobility using total FDI flows, defined as the sum of the absolute values of FDI net inflows and net outflows as a share of GDP. Both values are sourced from the IMF and World Bank via Our World in Data.                                                                                                                                                                                                                                                              | % of GDP                   | Z-Score + Min-max normalization to 0–10 per year                         | 5     |
| FDI Regulatory Restrictiveness Index | FDI restrictiveness is an OECD index gauging the restrictiveness of a country’s foreign direct investment (FDI) rules by looking at four main types of restrictions: foreign equity restrictions; discriminatory screening or approval mechanisms; restrictions on key foreign personnel and operational restrictions. We utilize the 2022 updated index methodology as published by the OECD, which is calculated from 2018 onward. This makes earlier year values incompatible with the newest values. | 0–1 (1 = most restrictive) | Z-Score + Min-max normalization to 0–10 per year + directionally flipped | 3.875 |
| Chinn-Ito Index                      | The Chinn-Ito index is an index measuring a country’s degree of capital account openness. We use their published ka_open variable which normalizes the index to 0-1.                                                                                                                                                                                                                                                                                                                                     | 0–1 (1 = most open)        | Z-Score + Min-max normalization to 0–10 per year                         | 4.025 |

### 9.3 Domain 3: Free Flow of Human Capital

| Dataset                          | Description                                                                                                                                                                    | Scale          | Transformation                                   | Grade |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------|-------|
| World Bank Net Migration         | Measures the difference between the number of immigrants and emigrants. Captures the overall balance of inward vs. outward migration.                                          | Net population | Z-Score + Min-max normalization to 0–10 per year | 4.25  |
| WEF Ease of Hiring Foreign Labor | Survey question: “In your country, how restrictive are regulations related to the hiring of foreign labor?” Measures openness of regulations and practices to foreign workers. | 1–7            | Z-Score + Min-max normalization to 0–10 per year | 4.075 |

## 10 World Economic Forum Executive Opinion Survey Methodology

### 10.1 The Executive Opinion Survey

The Executive Opinion Survey (the Survey) is the longest-running and most extensive survey of its kind. The Survey provides a yearly evaluation of critical aspects of socio-economic development for which statistical data is missing because it is extremely difficult to measure on a global scale at current statistical capacity. The aim of the Survey is to capture the particular reality of an economy, focusing on areas in which business leaders are well-positioned to provide an accurate evaluation. Through the survey questions, respondents are asked to evaluate their economy, complementing other statistical data to provide an assessment of the drivers of productivity and socio-economic progress.

The indicators derived from the Survey are used in the calculation of the Future of Growth Index in addition to a number of other World Economic Forum indexes, such as the Global Gender Gap Index and the Travel & Tourism Development Index; and several publications, including the Global Risks Report and the Energy Transition Report; as well as in many Forum activities and initiatives. A truly unique source of data, the Survey has also long been used by numerous international and non-governmental organizations, think tanks, and academia for empirical and policy work.

#### 10.1.1 Survey Administration

The 2024 edition of the Survey reached over 16,000 business executives between March and August 2024. Following the data review process described below, a total of over 11,000 responses were retained from 123 economies. More than 95% of the responses were collected in a digital-first way. The 2024 edition of the Survey was made available in 42 languages.

The latest edition of the Survey comprises 72 questions. The typical question asks respondents to evaluate their economy on a Likert scale of 1 (considered among the worst in the world) to 7 (considered among the best in the world). The questions are organized into eight topical areas: Infrastructure and Tourism; Dynamism and capacity of the private sector; Dynamism and capacity of the public sector; Enabling Environment; Talent and employment; Innovation Ecosystem; Managing Risks; Industrial Policy and Green Transition; and Artificial Intelligence.

The administration of the Survey is supervised by the World Economic Forum and conducted at the national level by the Forum's network of Partner Institutes. Partner Institutes are universities or research organizations, business associations, competitiveness councils, or in some cases market and social research companies. These organizations have the private-sector network to reach out to leading business executives, as well as a firm commitment to improving the socio-economic development of their respective economies.

The collection process is based on best practices in the field of survey administration. Partner Institutes are asked to follow detailed sampling guidelines to ensure the survey sample is representative of the economy in question and comparable across the globe. The sampling guidelines specify that Partner Institutes create a sample frame—a list of business executives from companies of various sizes and from the various sectors of activity. The sample frame should reflect the structure of the economy as follows:

- It should be in proportion to the share of GDP by sector: agriculture, manufacturing in-

dustry, non-manufacturing industry (mining and quarrying, electricity, gas and water supply, construction) and services.

- It should ensure the representation of both large- (more than 250 employees) and small-sized companies (250 employees or fewer), reflecting each sector. At least one-third of companies are large and one-third are small, with the remaining one-third determined by the structure of the economy in proportion to the share of GDP by company size.
- It should ensure that the chosen companies also have a sufficiently wide geographical coverage.
- It should ensure that respondents have sufficient seniority and experience to have a robust view on a variety of matters in their economy; in particular, respondents are currently employed at a position of senior manager or above.

The Survey is administered in a variety of formats. The majority of institutes distribute the survey through a digital mailing utilizing the Qualtrics Survey tool. A smaller share of the institutes conduct interviews over the phone or at respondents' place of work. To ensure higher levels of anonymity, interview-based in-field administration methods are only encouraged when economies have very low digitization but discouraged otherwise. GDPR-compliant contracting means institutes are legally bound by the high level of confidentiality.

### **10.1.2 Data Validation**

Respondent-level data is subjected to a careful review and validation process. The following exclusions are applied to respondent-level data:

- **Adequate survey completion:** Responses where completion of the questionnaire is less than 80% of the overall questionnaire, and the progress through the survey is less than 80%, are excluded.
- **Response fit with sampling criteria:** Respondents who do not fit seniority criteria are excluded.
- **Filter out test and duplicate responses:** Responses are checked to ensure there are no individuals who completed the survey twice or who completed the survey as a "test."
- **Identify and remove irregular responses:** A series of tests are used to spot responses where individuals did not pay adequate attention, or where they could have been completing the survey with false motives, by checking against speed, variance,  $z$ -tests, and answer matrix similarity.

In the 2024 edition, 150 economies were considered for inclusion, data was gathered for 129 economies, and 113 economies gathered sufficient data to be included in the EOS dataset. Among the economies not included in the dataset due to insufficient sample size were Albania, Barbados, Brunei Darussalam, Cape Verde, Israel, Kosovo, Lebanon, Liberia, Madagascar, Montenegro, Namibia, Nicaragua, the People's Republic of China, Slovakia, Trinidad and Tobago, and Zambia. Bahamas, Croatia, Iraq, and Uzbekistan gathered a successful dataset for the first year.

### **10.1.3 Computation of Country Scores**

Country scores are aggregated using a simple average of valid responses. All individual answers carry the same weight. Let  $q_{i,c,j}$  denote the answer to question  $i$  in country  $c$  from respondent

$j$ , and let  $N_{i,c}$  denote the number of respondents to question  $i$  in country  $c$ . The average Survey indicator  $i$  for country  $c$ , denoted  $q_{i,c}$ , is computed as:

$$q_{i,c} = \frac{\sum_{j=1}^{N_{i,c}} q_{i,c,j}}{N_{i,c}} \quad (1)$$

For each country and each Survey question, the final country score is a weighted average of scores based on two years of data collection. The weighted average approach serves three distinct purposes. First, it makes results less sensitive to the specific point in time when the Survey is administered. Second, it increases the amount of available information by providing a larger sample size. Third, because the Survey is typically carried out during the first half of the year, the average of the responses of the two editions better aligns the Survey data with many of the hard data indicators, which are typically annual.

The weighted scheme used to compute the final country score is composed of two overlapping elements. On the one hand, the calculation aims to give equal weight to each individual response across the two years, preventing a country's sample that is much larger in one year from overwhelming the smaller sample from the other year. On the other hand, the older sample is discounted to emphasise the most up-to-date information.

Let  $q_{i,c}^t$  denote economy  $c$ 's score for question  $i$  in year  $t$ , and let  $w_c^t$  denote its weight for that year. The two-year weighted score  $q_{i,c}^{2023-22}$  is then:

$$q_{i,c}^{2023-22} = w_c^{2022} \times q_{i,c}^{2022} + w_c^{2023} \times q_{i,c}^{2023} \quad (2)$$

The weights for each year are determined as follows, where  $N_c^t$  denotes the number of respondents for country  $c$  in year  $t$ :

$$w_c^{2022} = \frac{0.4 + \frac{N_c^{2022}}{N_c^{2022} + N_c^{2023}}}{2} \quad (3)$$

$$w_c^{2023} = \frac{0.6 + \frac{N_c^{2023}}{N_c^{2022} + N_c^{2023}}}{2} \quad (4)$$

In the case of Survey questions introduced in 2023 for which, by definition, no past data exists, full weight is given to the 2023 score. For countries whose 2023 data were discarded, results from the previous edition of the report are used instead.

As a final adjustment, some economies' scores are corrected for potential cultural bias. It has been evidenced that individuals in different countries use different score ranges in perception surveys, informed by different cultural norms—a phenomenon referred to as *acquiescence response bias*. The survey review process corrects for this bias by comparing the average country EOS-score level with the average “hard”-score level; EOS scores are then adjusted towards a common regression line against the comparative hard data.

## 11 Parabolic Scaling Methodology

The country set used to calibrate the parabolic transformations was selected through a structured review of the comparative capitalism literature and an evaluation against the eight institutional dimensions that define the GCI. The starting point was Bruno Amable’s typology, which identifies five varieties of capitalism (market-based, social democratic, Asian, Continental European, and Mediterranean) and assigns example countries to each system.<sup>1</sup> We retain Amable’s groupings as the backbone of the set: the United States, United Kingdom, Canada, and Australia for the market-based model; Denmark, Finland, and Sweden for the social democratic model; Japan and Korea for Asian capitalism; France, Germany, Belgium, Austria, Ireland, and Norway for Continental European capitalism; and Greece, Italy, Portugal, and Spain for the Mediterranean model.<sup>2</sup> We then extended the set to address the framework’s confinement to largely the Global North. Through focused regional reviews of Latin America, Sub-Saharan Africa, Eastern Europe, and Southeast Asia, we added Chile, Mauritius, and Estonia on the basis of their sustained alignment with the GCI’s institutional dimensions. Specifically, these countries demonstrate longstanding market liberalization, low average tariffs, private sector growth, and institutional frameworks oriented toward fair competition.<sup>3</sup> Singapore, South Africa, and Slovenia were considered but excluded because of persistent and substantial state participation in core market activity. Specifically, the prevalence of state holding companies in Singapore, racially targeted ownership transfer mandates in post-apartheid South Africa, and the almost 25 percent state ownership of the economy in Slovenia, all differ from the intention of this particular metric.<sup>4</sup>

The purpose of establishing this set is not to identify the “best” capitalist countries or to make a normative judgment about which countries are most capitalist. Rather, the set is intended to cover the empirical range of institutional configurations that the comparative capitalism literature, across multiple traditions, recognizes as capitalist. By averaging across this range to calibrate the parabolic peak for variables, the post-transformation data reflects what capitalist economies actually do in practice rather than what any single ideological tradition prescribes. This approach

<sup>1</sup>Bruno Amable, *The Diversity of Modern Capitalism* (Oxford: Oxford University Press, 2003).

<sup>2</sup>Amable’s typology builds on foundational work in the varieties of capitalism literature, particularly on Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001); J. Rogers Hollingsworth, Philippe C. Schmitter, and Wolfgang Streeck, eds., *Governing Capitalist Economies: Performance and Control of Economic Sectors* (New York: Oxford University Press, 1994); Colin Crouch and Wolfgang Streeck, eds., *Political Economy of Modern Capitalism: Mapping Convergence and Diversity* (London: Sage, 1997); and Richard Whitley, *Divergent Capitalisms: The Social Structuring and Change of Business Systems* (Oxford: Oxford University Press, 1999).

<sup>3</sup>On Chile’s post-1970s financial and trade liberalization, see Andrés Velasco, “Liberalization, Crisis, Intervention: The Chilean Financial System, 1975–1985,” IMF Working Paper WP/88/66 (Washington, DC: International Monetary Fund, 1988). On the institutional foundations of Mauritius’s growth model, see Jeffrey Frankel, “Mauritius: African Success Story” (working paper, Harvard Kennedy School, 2012); and Pritish Behuria, “The Political Economy of a Tax Haven: The Case of Mauritius,” *Review of International Political Economy* 30, no. 2 (2023): 772–800. On Estonia’s position within the post-communist varieties of capitalism literature, see Pavol Baboš, “Varieties of Capitalism in Central and Eastern Europe: Measuring the Co-Ordination Index of a National Economy,” *SEER: Journal for Labour and Social Affairs in Eastern Europe* 13, no. 4 (2010): 439–458.

<sup>4</sup>On Singapore, see Katalin Völgyi, “A Successful Model of State Capitalism: Singapore,” in *Seeking the Best Master: State Ownership in the Varieties of Capitalism*, ed. Miklós Szanyi (Budapest: Central European University Press, 2019), 275–296. On the South African race-based market economy, see Grietjie Verhoef, “‘Settlers and Comrades’. The Variety of Capitalism in South Africa, 1910–2016,” *Business History* 63, no. 8 (2021): 1413–1446. On Slovenia’s state ownership, see Dorothee Bohle and Béla Greskovits, “Neoliberalism, Embedded Neoliberalism, and Neocorporatism: Towards Transnational Capitalism in Central-Eastern Europe,” *West European Politics* 30, no. 3 (2007): 443–466; and Bertelsmann Stiftung, *BTI 2024 Country Report: Slovenia* (Gütersloh: Bertelsmann Stiftung, 2024).

preserves the institutional pluralism documented across the varieties of capitalism literature while still allowing the index to flag values that fall outside the range any recognized capitalist system has sustained. Critically, the calibration is applied only to variables for which there is strong theoretical reason to expect a nonlinear measurement relationship, wherein both very low and very high values are inconsistent with sustained market functioning. This process is only applied to two datasets in the current iteration of the GCI, but the process can be expanded to accommodate subsequent dataset additions when relevant.

Mechanically, we take this list of countries, gather their average scores for each year of a given data series, and compute the simple average of all those values so that each model receives equal weight. That aggregate average per year is then treated as the central value for the dataset during 0–10 scaling.